GS EXTERIOR EXPERTS · DECK DIVISION · PERMIT SET

5549 S Lansing Ct · Elevated Deck

5549 S Lansing Ct, Englewood · Rachel Chiu · 8' x 12' deck, 10' above the yard, off the existing door · TrueFrame 2x10 @ 16" on a (2)4x10 Douglas fir beam and (2) 6x6 DF #1 posts on Diamond Piers · TimberTech Reserve Driftwood, single picture frame, CONCEALoc · TimberTech Fulton rail, Black · posts + anchors for the owner-supplied square shade sail · October 19, 2026

SHEET INDEX

1. Elevated deck · **G-001** General notes · design criteria
2. Elevated deck · **C-100** Site plan
3. Elevated deck · **A-002** Isometric — assembled, labeled
4. Elevated deck · **A-003** Isometric — exploded, labeled
5. Elevated deck · **S-100** Foundation plan
6. Elevated deck · **S-101** Framing plan
7. Elevated deck · **S-102** Structure isometric, labeled
8. Elevated deck · **A-101** Decking plan
9. Elevated deck · **A-201** Railing plan + elevation
10. Elevated deck · **A-202** Elevations
11. Elevated deck · **A-301** Section + front elevation
12. Elevated deck · **D-501** Details
13. Elevated deck · **D-502** Connection details — frame
14. Elevated deck · **D-504** Connection details — deck edge, rail, lateral

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale. This set carries no pricing.

G-001 GENERAL NOTES · DESIGN CRITERIA

Loads, codes, structure summary, sheet index, site — the permit-set cover

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PROJECT

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech Reserve Driftwood, Fulton black, shade sail)
5549 S Lansing Ct, Englewood CO 80111
Prepared for Rachel Chiu
92 SF · 10'-0" above grade · 1 zone

DESIGN CRITERIA

Deck live load	40 psf uniform (IRC Table R301.5); guards 200 lb concentrated / 50 plf
Ground snow	30 psf (ground; deck design uses the larger of snow and live) — 40 psf live governs; 30 psf ground snow, roof 30 psf non-reducible (Arapahoe County design criteria); deck tables at 40 psf live + 10 psf dead
Dead load	10 psf deck
Roof snow (cover)	—
Design total	50 psf on the deck
Wind	115 mph Vult (3-second gust), Exposure B (suburban), Risk Cat. II — county design-criteria table (2015 cycle; confirm the 2021 table)
Frost depth	3'-0" (footings bear below)
Soil bearing	1,500 psf presumptive (IRC R401.4.1) — no soils report
Seismic	B
Wildfire (WUI)	not in a WUI zone
Codes	2021 IRC / IBC as adopted by Arapahoe County (Res. 21-394, eff. 4/1/2022); IRC R507 decks with the county-amended Colorado ICC Deck Building Guide; ASCE 7-16 loads; Tables R507.5 / R507.6 / R507.4 / R507.9
Engineering	Stamped structural set by a Colorado PE required — the stamped set governs

STRUCTURE

Footings	DP-50/50
Posts	2 x 6x6 Douglas fir #1 with 4x4 knee braces
Beams	FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)
Joists	7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"
Rims	2-ply 2x10 sides; front rim = flush beam
Ledger	2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners
Decking	TimberTech Reserve Sand — 24 rows parallel to the house, picture frame in Reserve Driftwood, 3/16" gaps, CONCEALoc
Rail	Fulton 36" Black: 5 bays, 6 posts, 31 LF

SHEET INDEX

G-001	General notes · design criteria
C-100	Site plan
A-001	Isometric · renders
A-002	Isometric — assembled, labeled
A-003	Isometric — exploded, labeled
S-100	Foundation plan
S-101	Framing plan
S-102	Structure isometric, labeled
A-101	Decking plan
A-201	Railing plan + elevation
A-202	Elevations
A-301	Section + front elevation
D-501	Details

SITE

Rear / front lot line: deck edge 13.0' from the line, setback per the PUD...
Side lot line: deck edge 23.7' from the line, 5' required
Easement: Check plat R1535678 for rear / side utility and drainage...
Unincorporated Arapahoe County (Cherry Creek CDP; Englewood mailing address). Parcel 2075-14-4-05-001 / PIN 031772427, Lot 41 Blk 9 Cherry Creek Vista 6th Flg Amend Resub (plat R1535678), 6,401 SF corner lot (front S Lansing Ct east, street side E Berry Ave north). R-P PUD Cherry Creek Vista (Z75-030 / P75-027 / P75-028). Deck on the WEST (rear) wall, north end, hung on the 2' second-floor cantilever that runs the full 33' wall (assessor sketch). Tax: CO 2.9 + RTD 1.0 + CD 0.1 + Arapahoe 0.25 = 4.25% (no city).

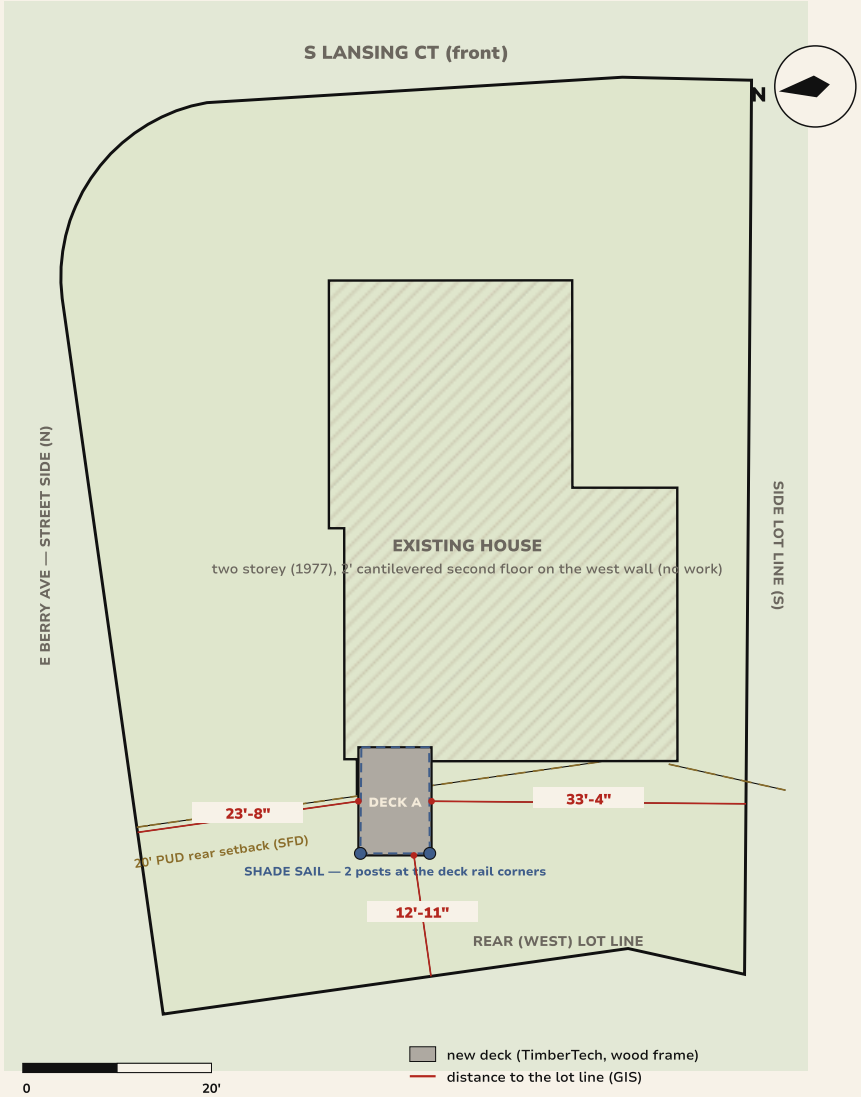
Elevated deck · C-100 · Site plan

C-100 **SITE PLAN**

Parcel, existing house, the new deck, distances to the lot lines, setbacks

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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PARCEL

5549 S Lansing Ct, Englewood CO 80111
Parcel 2075-14-4-05-001 · PIN 031772427
Lot 41, Block 9, Cherry Creek Vista 6th Filing Amended Resub (plat R1535678)
Unincorporated Arapahoe County · zoning R-P PUD (Z75-030 / P75-027 / P75-028)
Lot 6,401 SF (GIS) · corner lot · flood zone X
House 1977, 2 storey, 1,704 SF + 660 SF basement + 441 SF garage (assessor)
HOA: Cherry Creek Vista II

LOT COVERAGE

Lot area	6,401 SF
Lot coverage limit	none in the PUD plan
Existing deck (assessor)	48 SF (8' x 6') — removed
New deck	8' x 12' · ~92 SF of frame, 10' above grade

SETBACKS

Elevated decks get no setback projection allowance (Arapahoe LDC) — the deck must meet the setbacks
GIS puts the new deck front ~13' off the west line: passes the 10' standard, not the 20' — confirm the governing column and the rear line (corner lot) with Zoning before submittal
E Berry Ave street side 10' required — ~23'-8" clear; south side 5' — ~33' clear
Measured (red): to the nearest lot line from the new work — GIS, not survey

SITE WORK

· Remove the existing 8' x 6' deck and its two posts / pads; haul off
· Two Diamond Pier DP-50 foundations (no excavation spoils), 6x6 DF #1 posts with 4x4 knee braces
· New ledger on the 2' cantilevered second floor per the engineer's detail; membrane + Z-flash, (4) DTT1Z lateral ties
· No stairs — the deck is reached from the house door only
· Call 811 before the piers; protect the lawn and the patio slab at the north wall

Arapahoe County GIS (address point, parcel, 2024 roof prints, R-P zoning, jurisdiction, flood, wildfire layers), county assessor record and sketch, 2nd Amended PDP P75-027 setback table, county deck permit guide (rev. 8/20/2024) and the county-amended ICC deck guide. GIS lines are not a survey — an ILC governs if Zoning asks.

Plan · approx. 1" = 20' at full size · GIS lines, not a survey

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-002 · Isometric — assembled, labeled

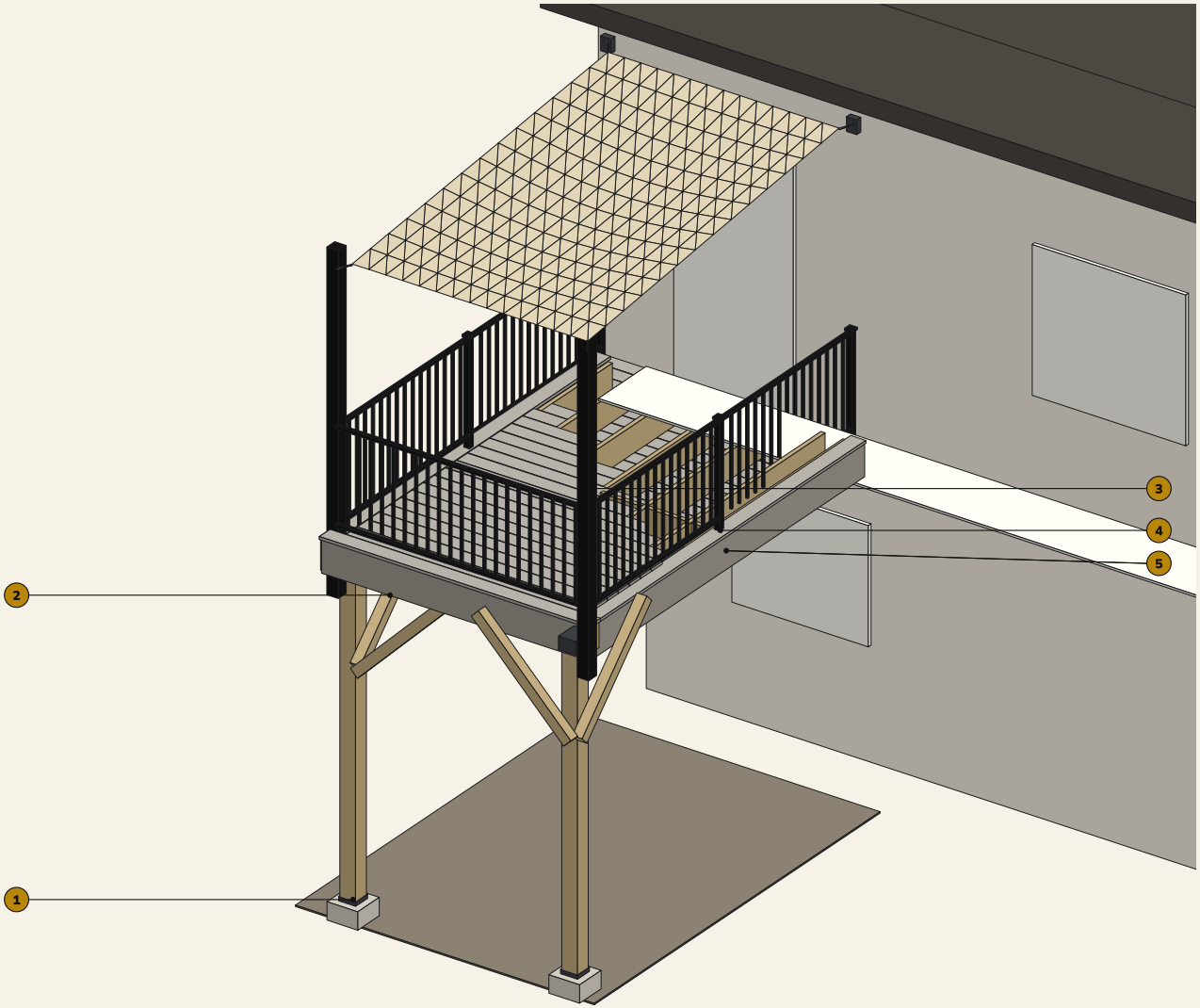
A-002

ISOMETRIC — ASSEMBLED, LABELED

Everything you see called out: piers, decking, picture frame, skirt, stair, roof covers, finishes

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
3. TimberTech Reserve Driftwood field decking, hidden fasteners
4. Picture frame + breaker boards — TimberTech Reserve Driftwood
5. Fascia

House shown for context only (the deck work only). The frame under the decking is called out on S-102 and A-003.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-003 · Isometric — exploded, labeled

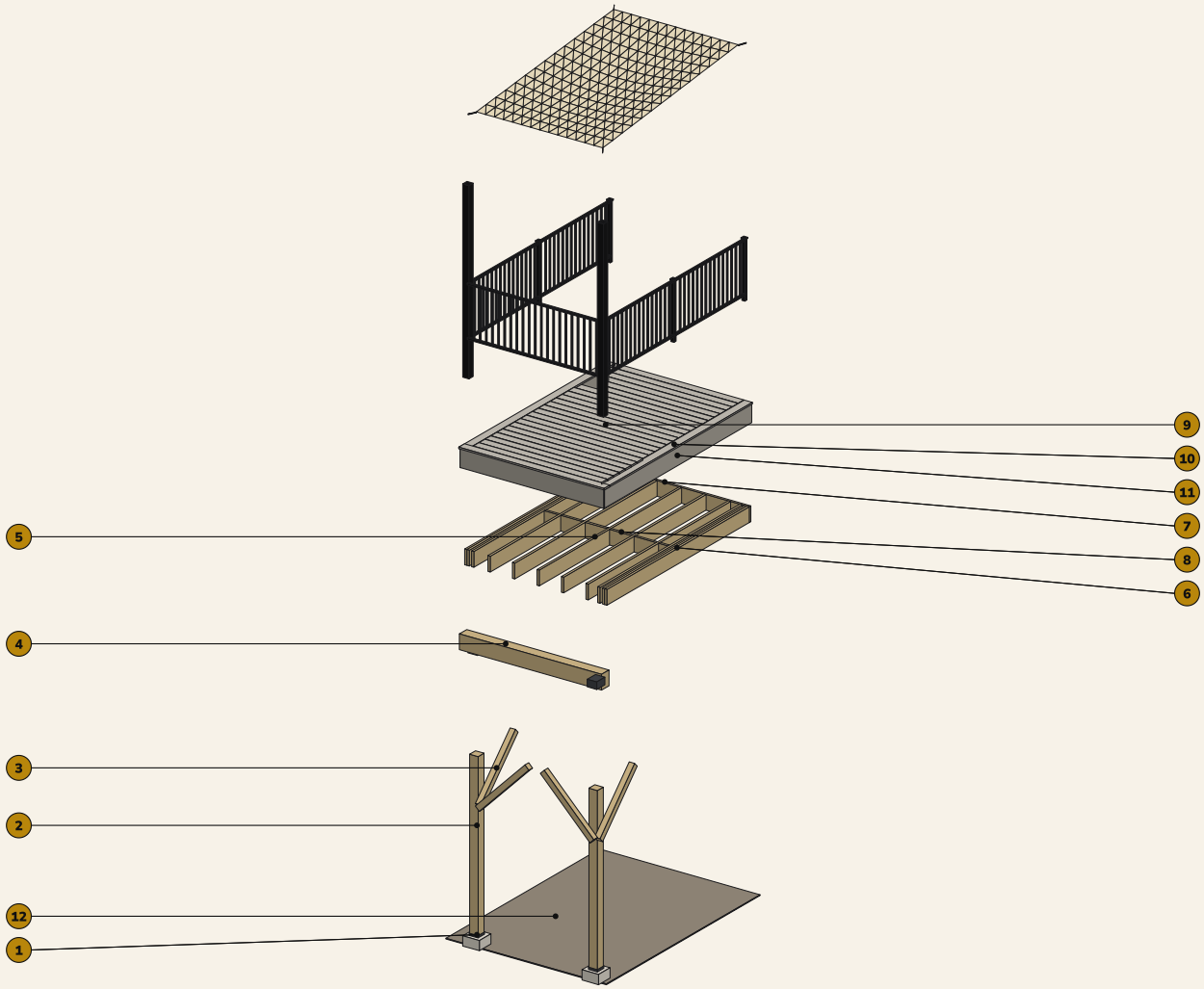
A-003

ISOMETRIC — EXPLODED, LABELED

Each build layer lifted in order: piers · posts + knee braces · beams · frame · decking · rail

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 · PERMIT SET



KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. TimberTech Reserve Driftwood field decking, hidden fasteners
10. Picture frame + breaker boards — TimberTech Reserve Driftwood
11. Fascia
12. Gravel over 6-mil poly under the decks

Layers lifted in build order. Lifts are for clarity only; every member returns straight down to its place.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · S-100 · Foundation plan

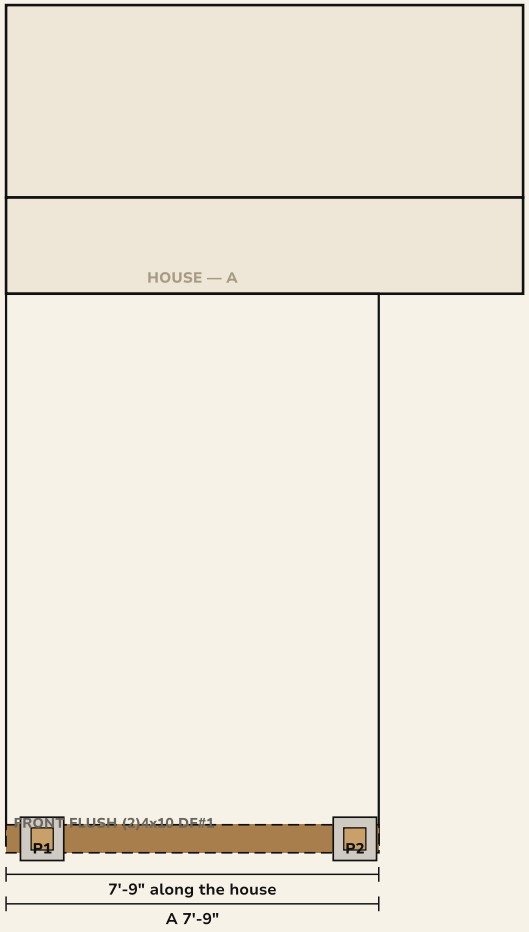
S-100

FOUNDATION PLAN

Footings, posts and beam lines with the footing schedule; beams dashed (below the joists)

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 · PERMIT SET



FOOTING SCHEDULE

Diamond Pier DP-50/50 (head + 4 x 50" pins), 3,300 lb allowable

ID	X	Y	ZONE	LOAD
P1	9"	11'-4"	A	1,821 lb
P2	7'-3"	11'-4"	A	1,821 lb

x from the left frame face facing the house; y from the reference wall.
Frost 3'-0" · soil 1,500 psf · worst post 1,821 lb

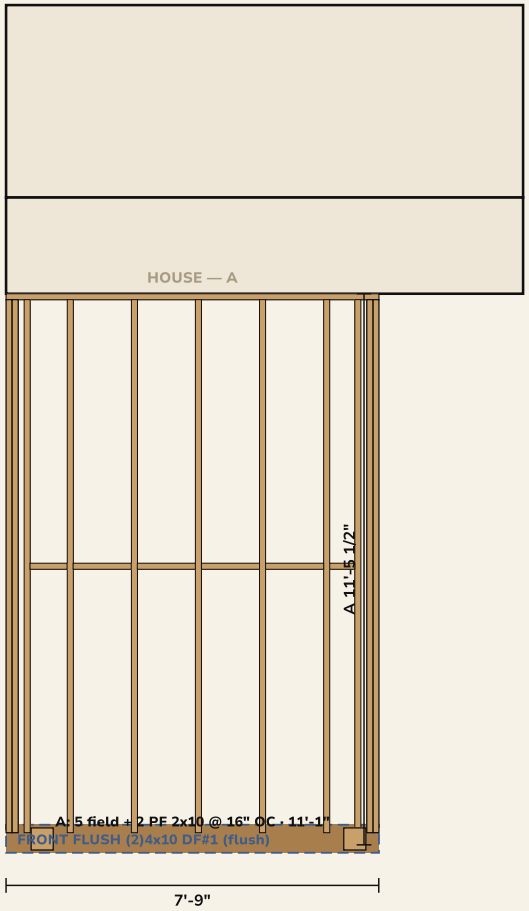
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

S-101 FRAMING PLAN

Ledger, joists, rims, blocking, beams; every member in the model — the cut list is on T-600

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRAMING NOTES

7 2x10 #1 True Frame Joist SYP GC @ 16" OC x 11'-1"

2-ply 2x10 sides; front rim = flush beam

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

blocking one row over the beam, staggered

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)

Connectors:

1/2" x 10" HDG hex bolt + nut + 2 washers — knee braces (brace to the 6x6 post): 4

1/2" x 12" HDG hex bolt + nut + 2 washers — knee braces (brace to the (2)4x beam): 2

1/2" x 6" HDG hex bolt + nut + 2 washers — knee braces (brace to the joist block): 2

Simpson LUS28Z face-mount hanger 2x8/2x10: 14

Simpson HUCQ210-2-SDS concealed-flange double 2x10 hanger (SDS screws included): 4

Simpson DTT1Z lateral tie kit w/ SDS screws: 4

Simpson CCQ column cap, 7" seat for the (2)4x10 beam on a 6x6 post (SDS screws incl.) — confirm SKU / engineer's equal:

1/2" x 8" HDG carriage bolt + nut + washer: 10

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · S-102 · Structure isometric, labeled

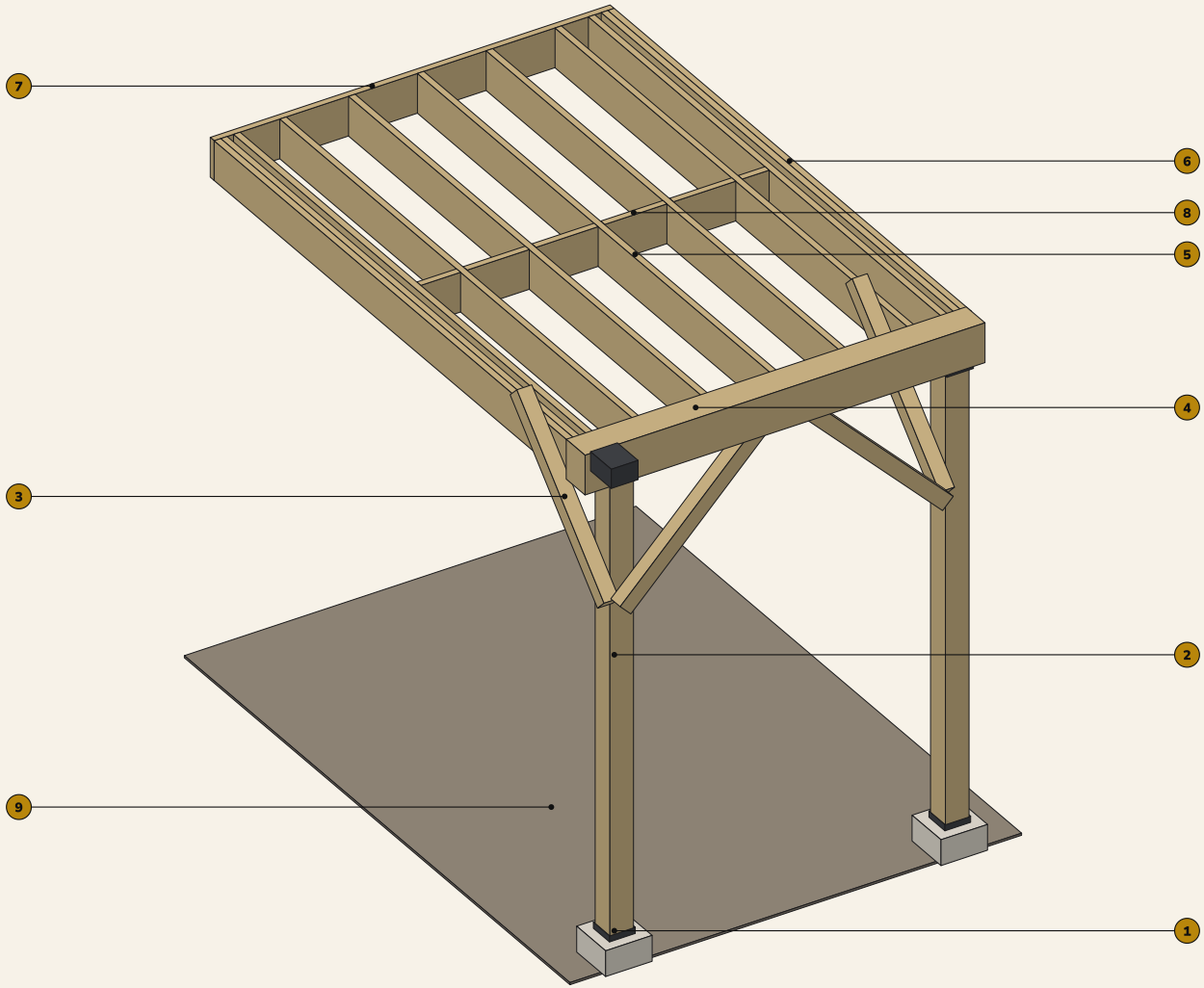
S-102

STRUCTURE ISOMETRIC — FRAME, LABELED

Piers, post bases, posts, knee braces, post caps, beam, joists, double rims, ledger, blocking

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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KEY

1. Diamond Pier DP-50/50 — head + 4 driven pins (frost-protected, no excavation)
2. 6x6 Douglas fir #1 post on a ABA66Z standoff base
3. 4x4 DF #1 knee brace at 45°, both directions, through-bolted
4. Beam
5. 2x10 joist
6. Rim
7. Ledger
8. Blocking
9. Gravel over 6-mil poly under the decks

Decking, rail and house removed to show the frame. Joists hang from the ledger in face-mount hangers, bear on the drop beam with H2.5AZ ties; double rims in concealed-flange hangers.

Axonometric · not to scale · the model the plans are drawn from

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-101 · Decking plan

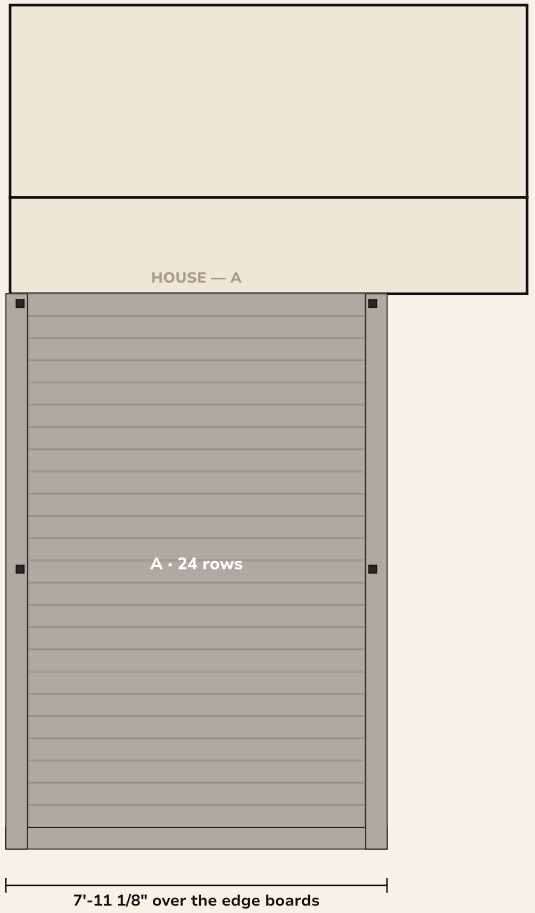
A-101

DECKING PLAN

Every board, border and divider as it lays; rail posts shown for the pockets

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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DECKING

Field: TimberTech Reserve Driftwood, 5.36" x 0.94", 3/16" gaps

Border: Reserve Driftwood

Boards run parallel to the house; edge overhang 1 1/16" over the fascia

CONCEALoc at every joist in every gap (168)

Cap-Tor xd color-matched ONLY at: first row 1 per joist on the house edge · borders 2 per bearing point @ 16" · fascia 2 at each board end then

Rows per zone: A 24 (no rips)

Plan · not to scale · house at the top · dimensions govern

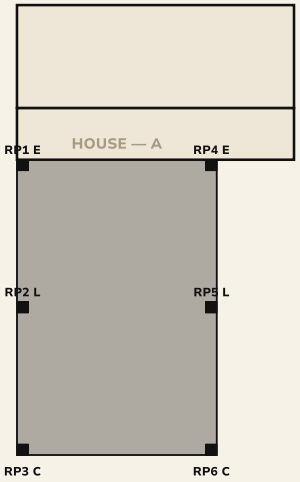
Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

A-201 RAILING PLAN + ELEVATION

Post IDs and types (E end · L line · C corner), bay lengths, outer-edge elevation

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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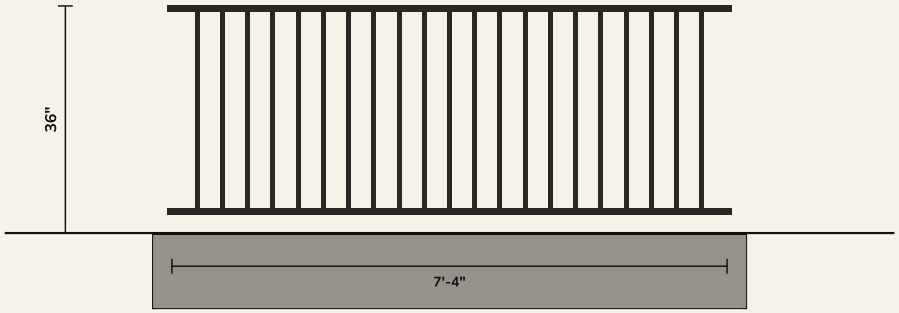
RAILING

Fulton 36" Black: 5 bays, 6 posts, 31 LF

Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

Bays: left 5'-6 1/4" · left 5'-6 1/4" · right 5'-6 1/4" · right 5'-6 1/4" · front 7'-4"

Guard 36" min above the deck; openings under 4" (IRC R312). Rail LF for labor = deck-edge LF of actual rail.



Plan above · elevation of the outer edge below

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · A-202 · Elevations

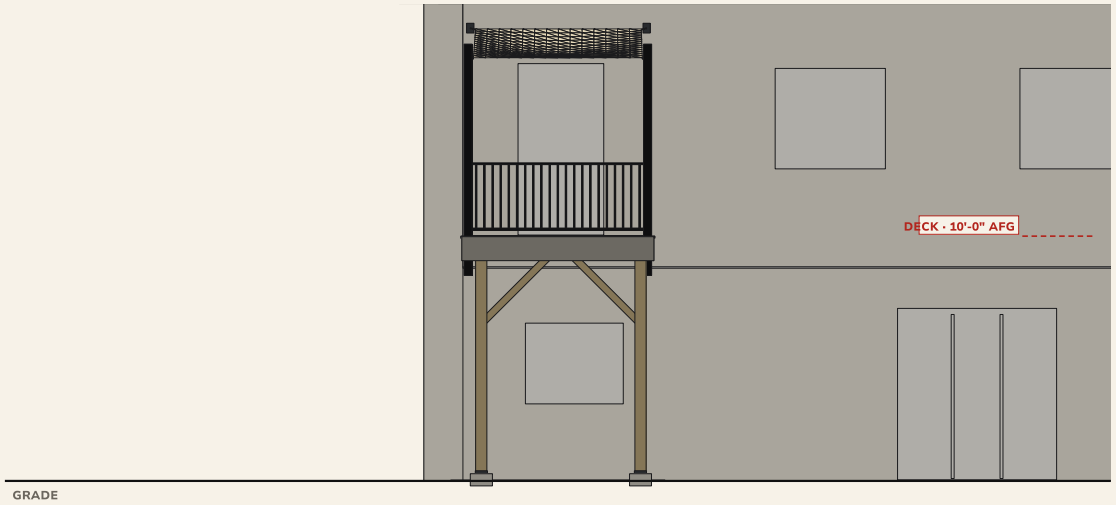
A-202 **ELEVATIONS**

Front from the yard and both ends, drawn from the model — deck and roof-cover heights

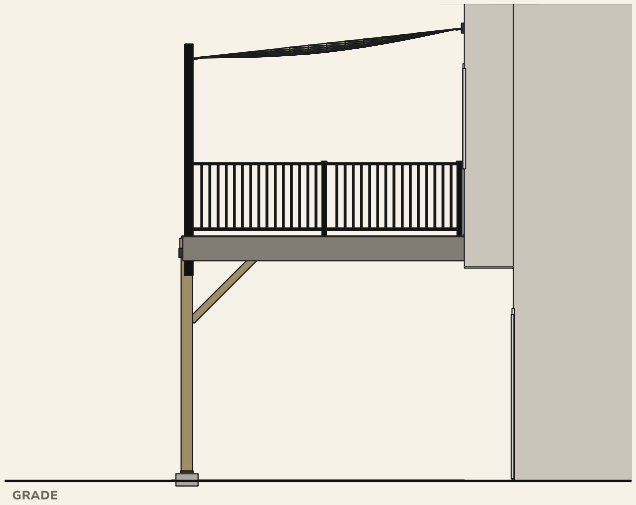
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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FRONT ELEVATION — from the yard



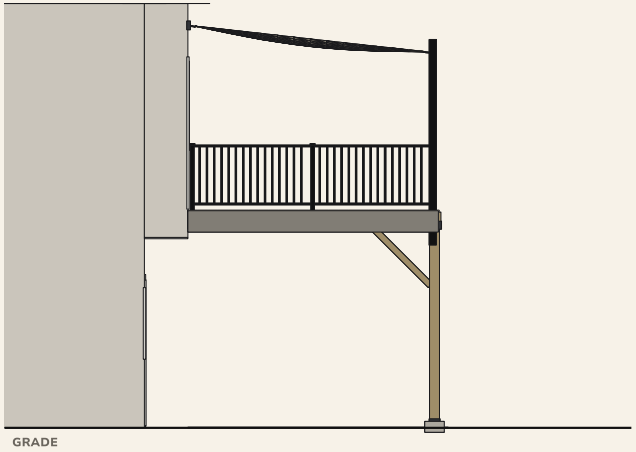
RIGHT END ELEVATION



NOTES

· Deck surface 10'-0" above grade
· Heights AFG = above finished grade at the beam line; field-verify grade before setting the piers
· Deck frame: 2x10 #1 True Frame GC @ 16" OC on a (2)4x10 DF #1 drop beam, 2 posts with 4x4 knee braces both ways
· Guard 36" min. on every open side — the surface is 10'-0" above grade (IRC R312.1)

LEFT END ELEVATION



Elevations · orthographic · not to scale · heights govern

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

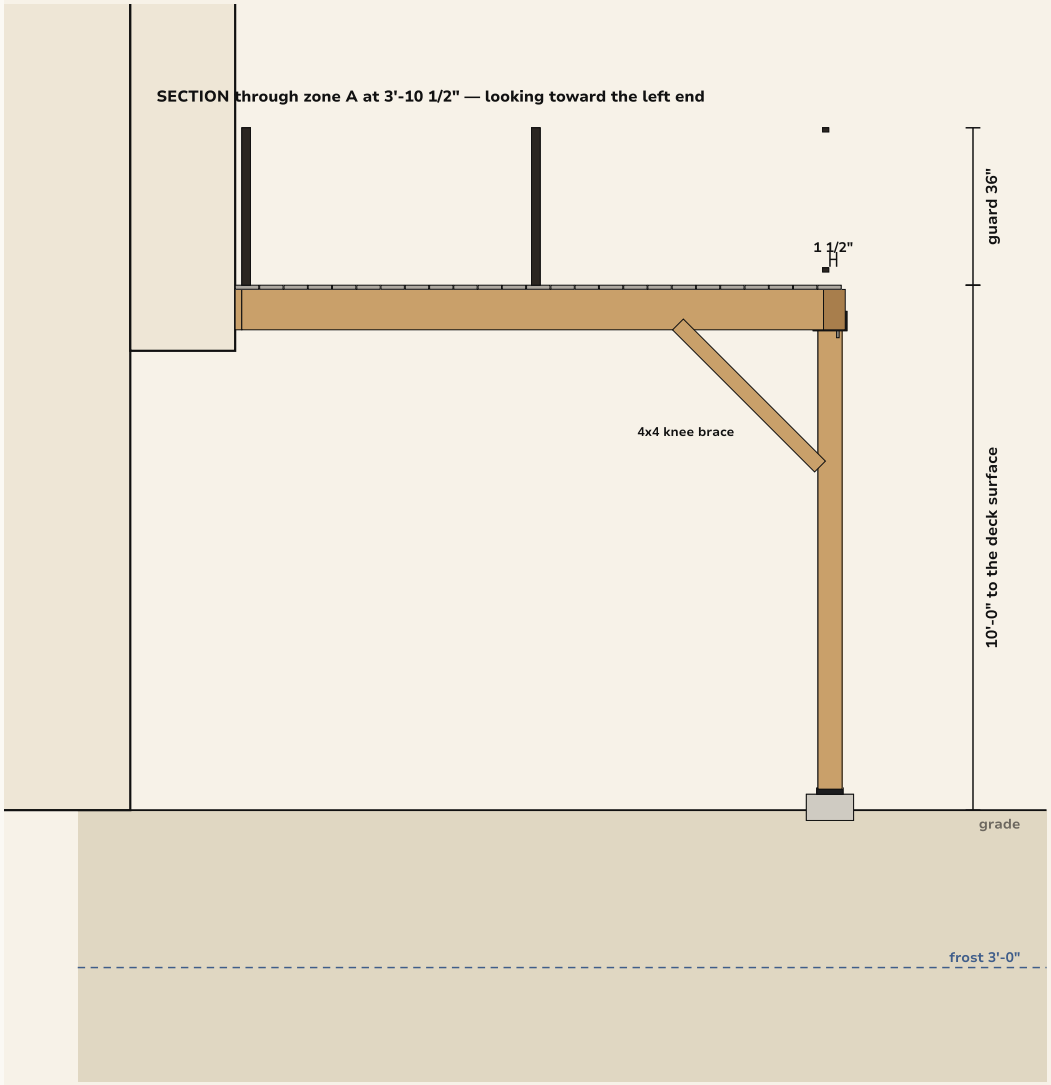
A-301 SECTION + FRONT ELEVATION

Cut through the deepest zone; heights to grade, frost line, cantilever, guard

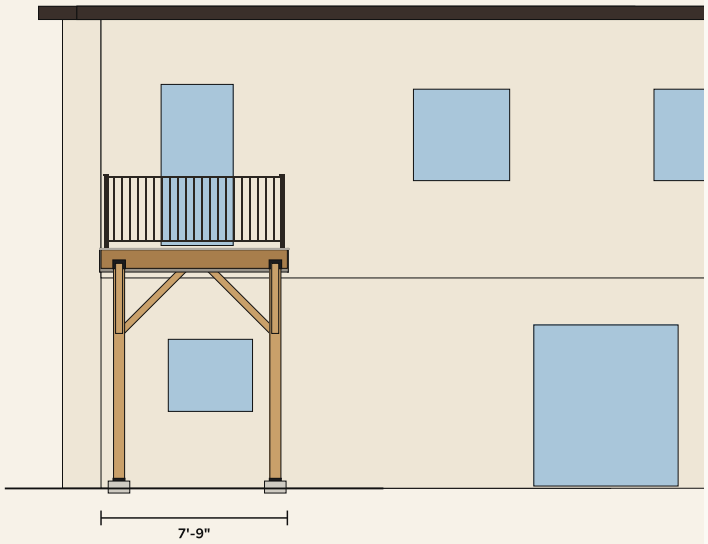
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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SECTION through zone A at 3'-10 1/2" — looking toward the left end



FRONT ELEVATION — from the yard



SECTION NOTES

Deck surface 10'-0" above grade; guard 36" where the drop exceeds 30"

FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)

Posts 2 x 6x6 Douglas fir #1 with 4x4 knee braces on DP-50/50

2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

G-Tape 2" every joist, PF joist, blocking and the ledger top · 4" on both rims and the beam(s)

Section left · elevation right · not to a common scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

Elevated deck · D-501 · Details

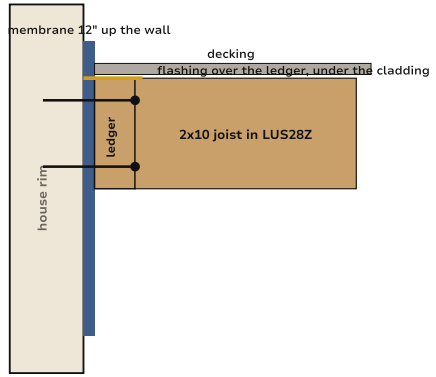
D-501 **DETAILS**

Ledger · post / beam / cap · footing · rail post · deck edge · the job's special condition

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

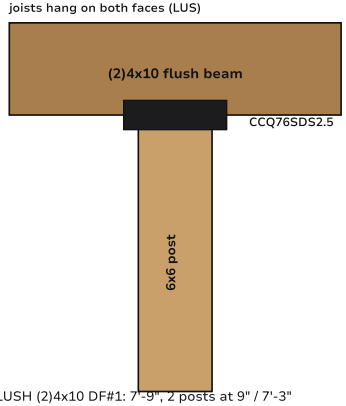
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1 LEDGER AT THE HOUSE



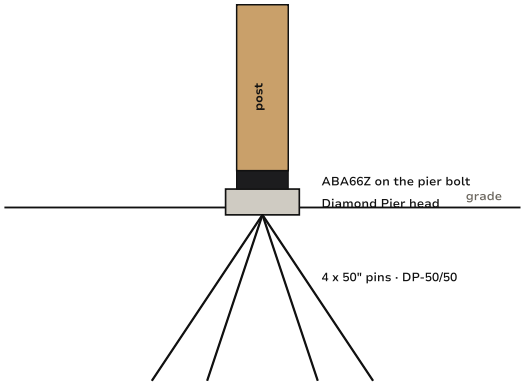
2x10 ledger x 7'-9" total · LedgerLOK 2 rows 2" from top/bottom, staggered 12" OC, first 3" from each end · 9 fasteners

2 POST · BEAM · CAP

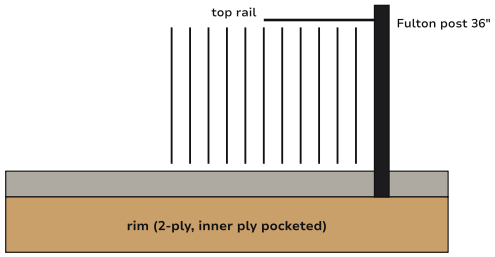


FRONT FLUSH (2)4x10 DF#1: 7'-9", 2 posts at 9" / 7'-3" (worst 1,821 lb)

3 FOOTING · POST BASE

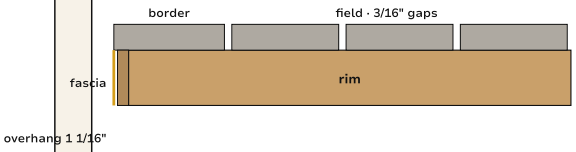


4 RAIL POST · GUARD



Fulton 36" — 6 posts (2 CORNER, 2 END, 2 LINE) inside the outer rim ply, inner ply pocketed 2" wide, (2) 7/16" x 4-1/2" bolts per post

5 DECK EDGE · BORDER



CONCEALoc at every joist in every gap (168)

6 FLUSH BEAM

Flush beam: top of beam = top of joists. Joists hang on both faces in LUS hangers; opposing hangers per Simpson and the stamped set.

Schematic details, not to scale

Design intent. The stamped engineering set governs structure, dimensions and locations. Field verify. Do not scale.

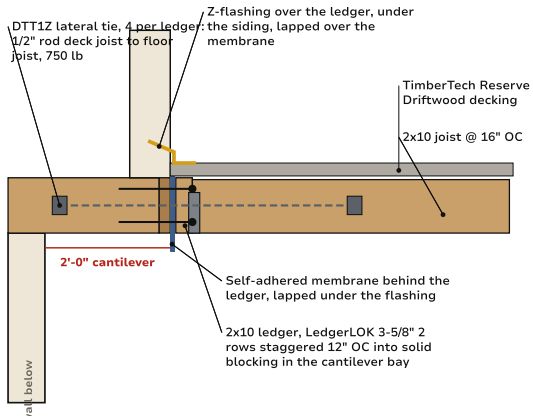
D-502 CONNECTION DETAILS — FRAME

Ledger on the cantilever · hangers · joists on the beam · (2)4x10 on the cap · pier + base · knee braces

5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

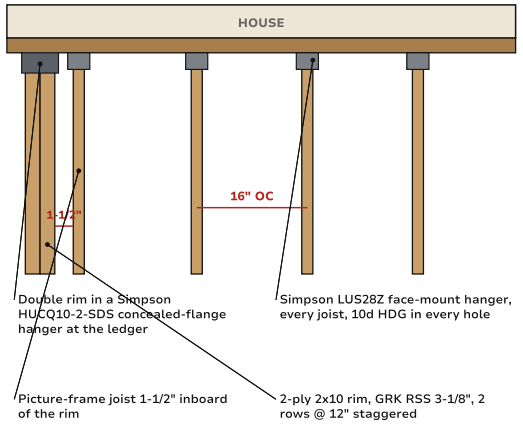
GSX · Jade Pender 303-550-9558 · October 19, 2026 · Rev 0 · PERMIT SET

1 LEDGER ON THE CANTILEVERED FLOOR — SECTION

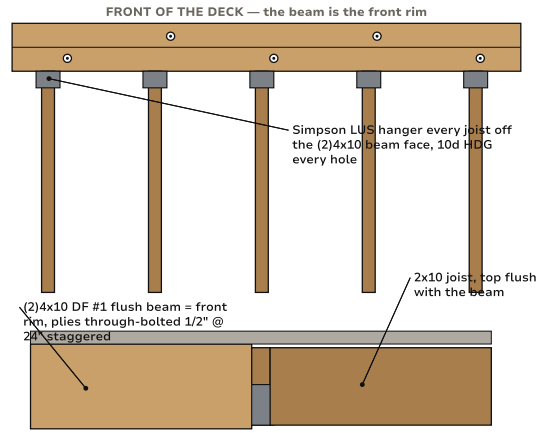


ENGINEER: Arapahoe County bars a ledger on a cantilevered floor without engineer approval. The stamped detail governs: ledger into the blocked cantilever joists, or a freestanding beam on posts at the house.

2 HANGERS — PLAN AT THE LEDGER + SIDE RIM

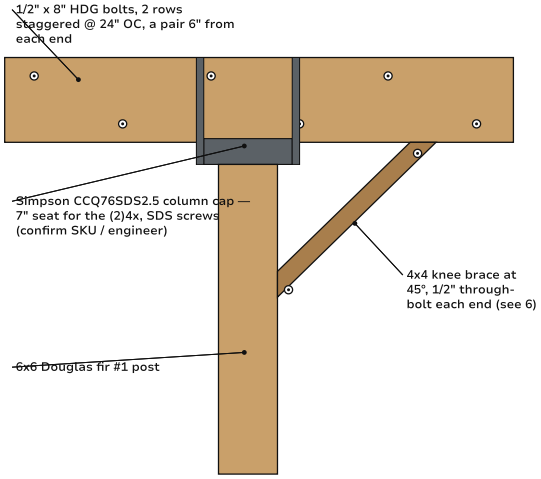


3 JOISTS INTO THE FLUSH BEAM — PLAN + SECTION

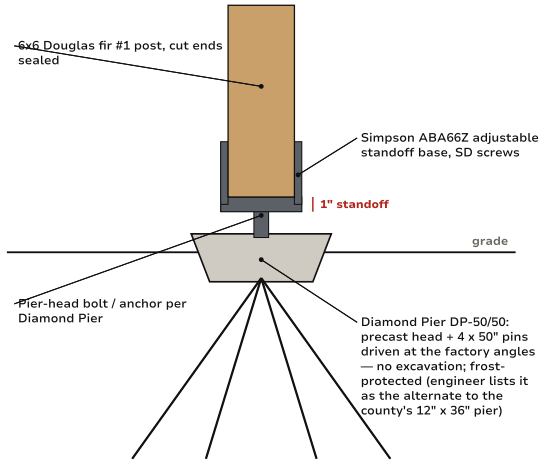


No cantilever: the beam carries the front edge, joists hang both ends

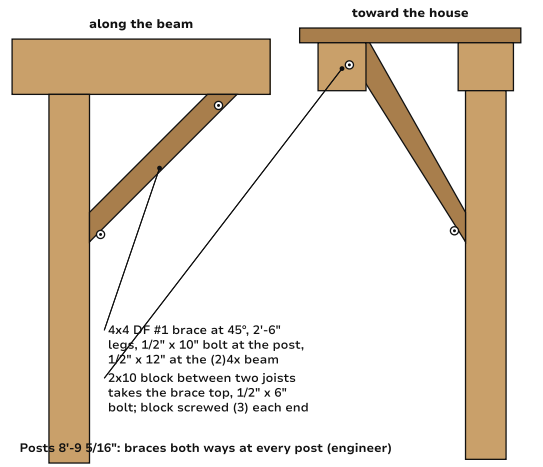
4 (2)4x10 BEAM ON THE 6x6 — ELEVATION



5 POST BASE ON THE DIAMOND PIER — SECTION



6 KNEE BRACES — BOTH DIRECTIONS



Posts 8'-9 5/16": braces both ways at every post (engineer)

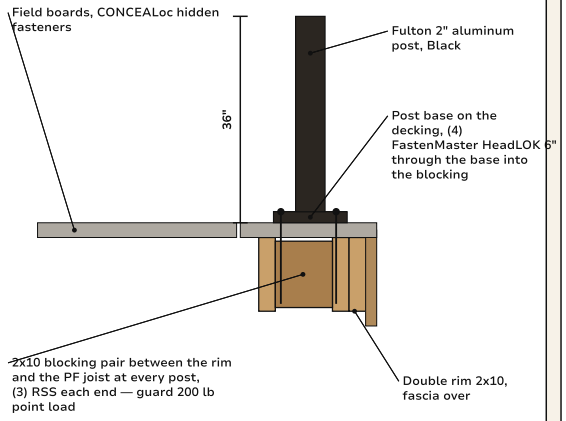
D-504 CONNECTION DETAILS — DECK EDGE, RAIL, LATERAL

Rail post mounting · picture frame + fascia · guard bay · DTT1Z lateral ties · joist tape · fastening schedule

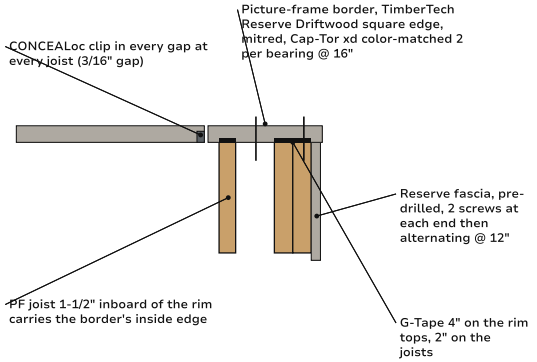
5549 S Lansing Ct — elevated deck, 8' x 12' at 10' (TimberTech...

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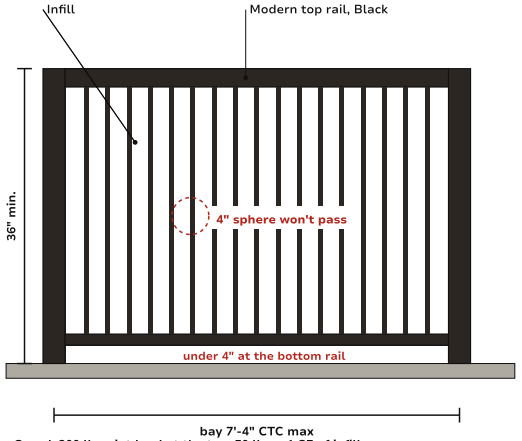
1 RAIL POST AT THE RIM



2 DECK EDGE — PICTURE FRAME + FASCIA

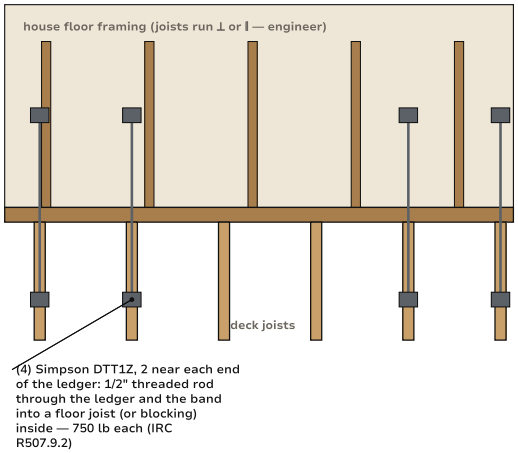


3 GUARD — ONE BAY, ELEVATION

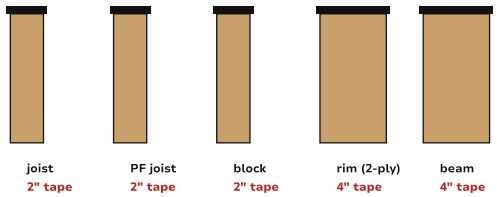


Guard: 200 lb point load at the top, 50 lb on 1 SF of infill (IRC Table R301.5) — submit the rail code report

4 LATERAL LOAD — DTT1Z TIES (PLAN)



5 JOIST TAPE + BLOCKING

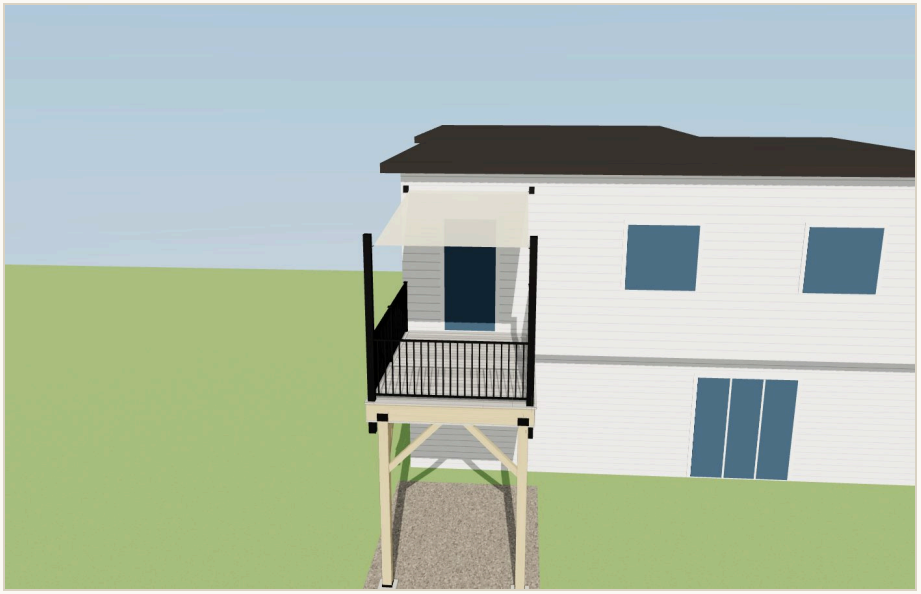


Nichigo G-Tape 3035 (black) on every top edge before decking, lapped over the ends and down the sides 1/2"; one row of 2x10 blocking over the beam, (3) structural screws each

6 FASTENING SCHEDULE

Ledger to house	LedgerLOK 3-5/8", 2 rows staggered 12" OC — engineer
Lateral	(4) DTT1Z, 1/2" rod into the floor framing
Joist to ledger	LUS28Z, 10d HDG every hole
Rim to ledger	HUCQ10-2-SDS, SDS screws
Rim plies	GRK RSS 3-1/8", 2 rows @ 12" staggered
Joist to beam	H2.5AZ each joist + rim, 10d x 1-1/2"
Beam plies	1/2" x 8" HDG bolts, 2 rows staggered @ 24"
Beam to post	CCQ76SDS2.5 cap, SDS screws
Post to pier	ABA66Z on the Diamond Pier, SD screws
Knee braces	1/2" through-bolt each end (x10 / x12 / x6)
Field boards	CONCEALoc, every joist, every gap
Borders / fascia	Cap-Tor xd color-matched, pre-drilled
Rail posts	(2) 7/16" bolts per post

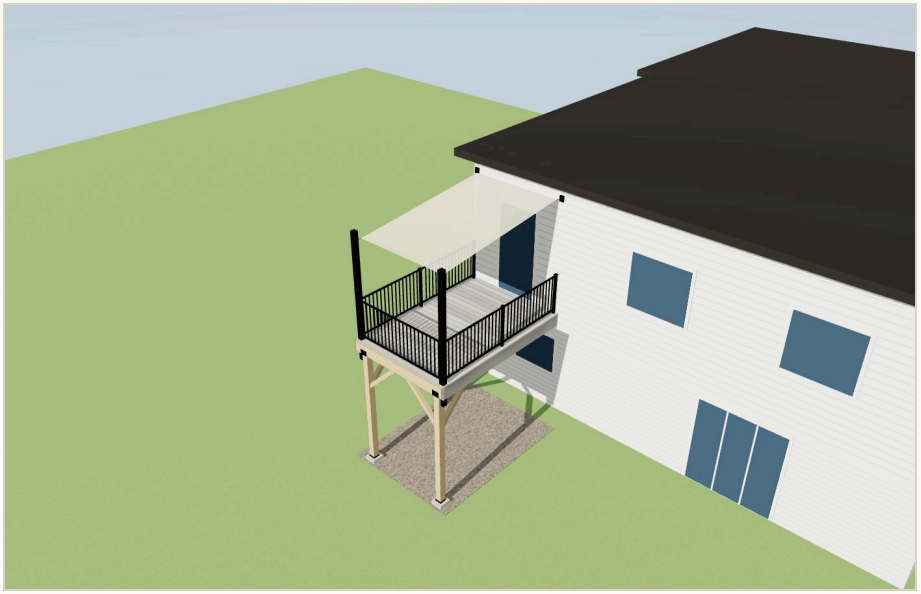
Deck



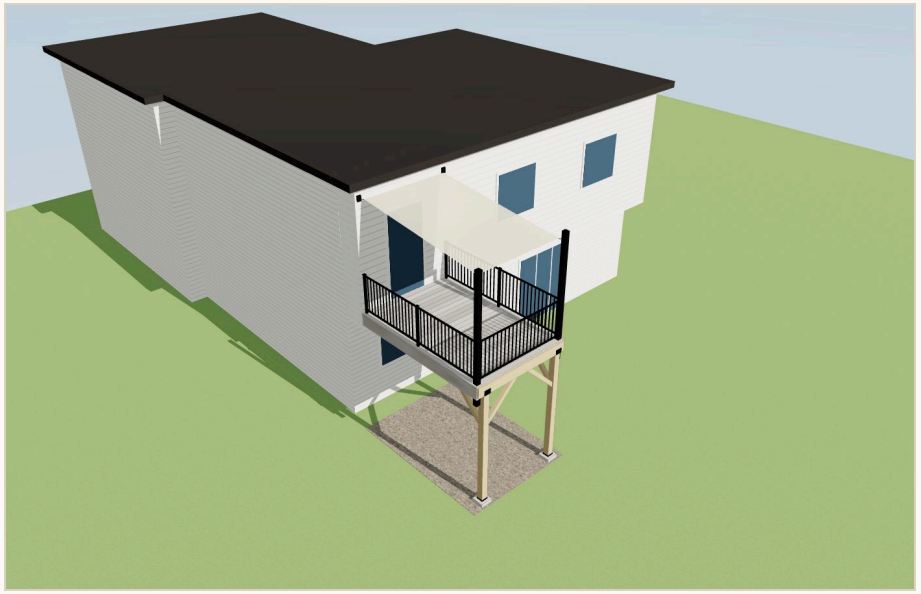
From the yard



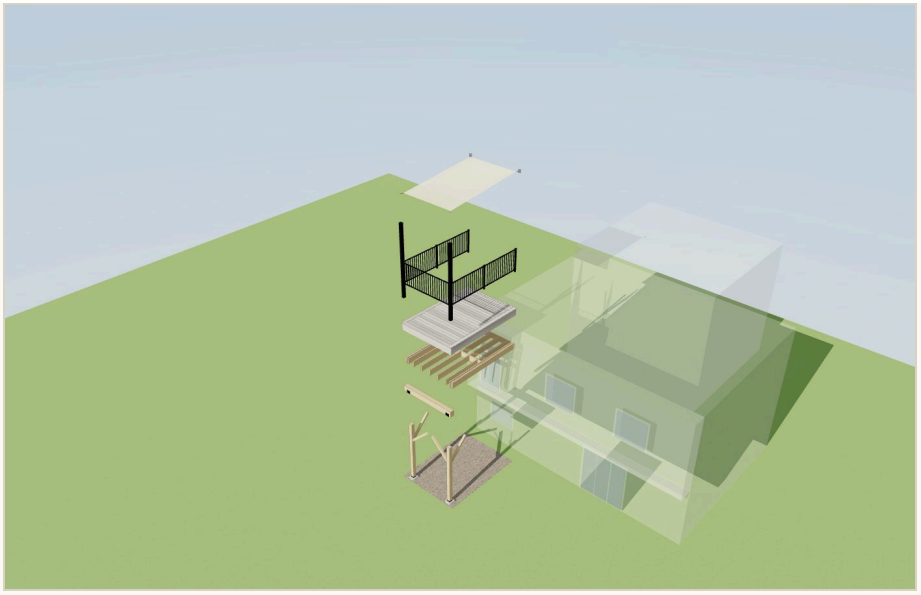
From the front corner



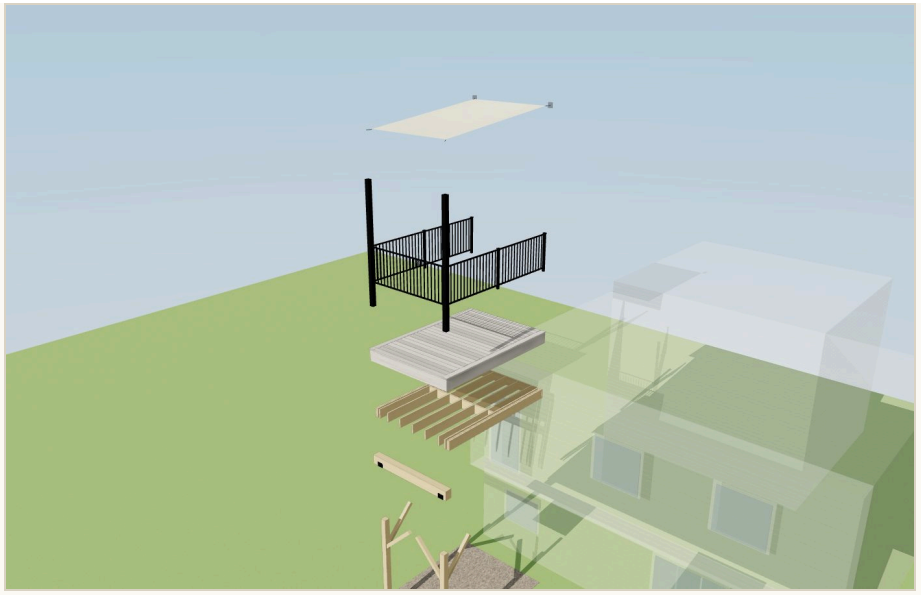
Isometric



Isometric — opposite corner



Exploded — every layer



Exploded — from the corner



Structure from below



Under the deck